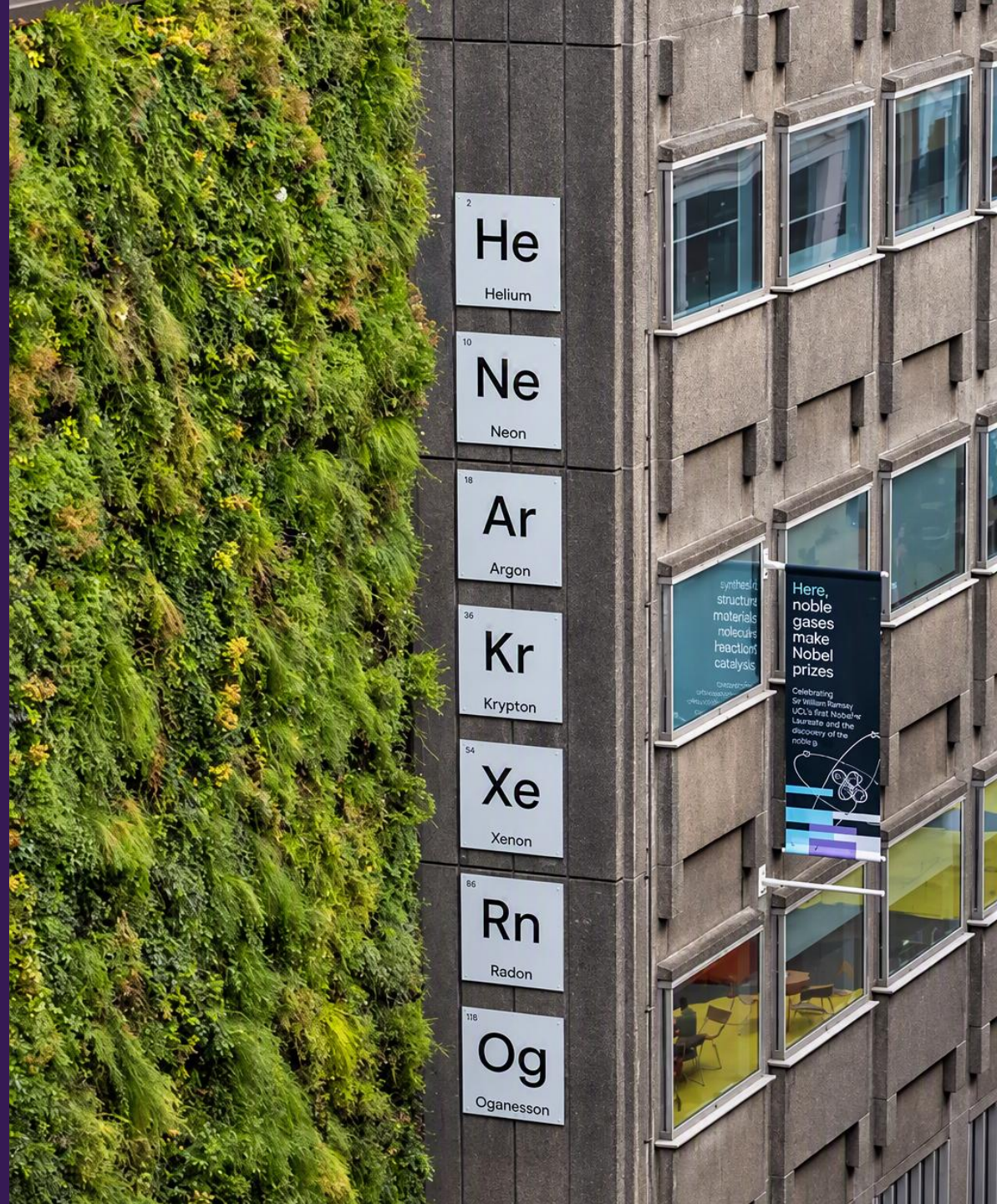
Thermodynamics 1



Dr Sam Cassidy



The Rules

The First Rule

Energy can neither be created nor destroyed, only converted from one form to another

The Second Rule

The entropy of an isolated system always increases over time

The Third Rule

As a system approaches absolute zero temperature, its entropy approaches a constant minimum value

The Fourth Rule

If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other

The Laws

The Zeroth Law

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The Laws

Zeroth Law

If two systems are in thermal equilibrium with a third system, they are in thermal equilibrium with each other

A=B B=C C=A





The Laws

The First Law

Energy can neither be created nor destroyed, only converted from one form to another



The Laws



The Second Law

The entropy of an isolated system always increases over time



The Laws



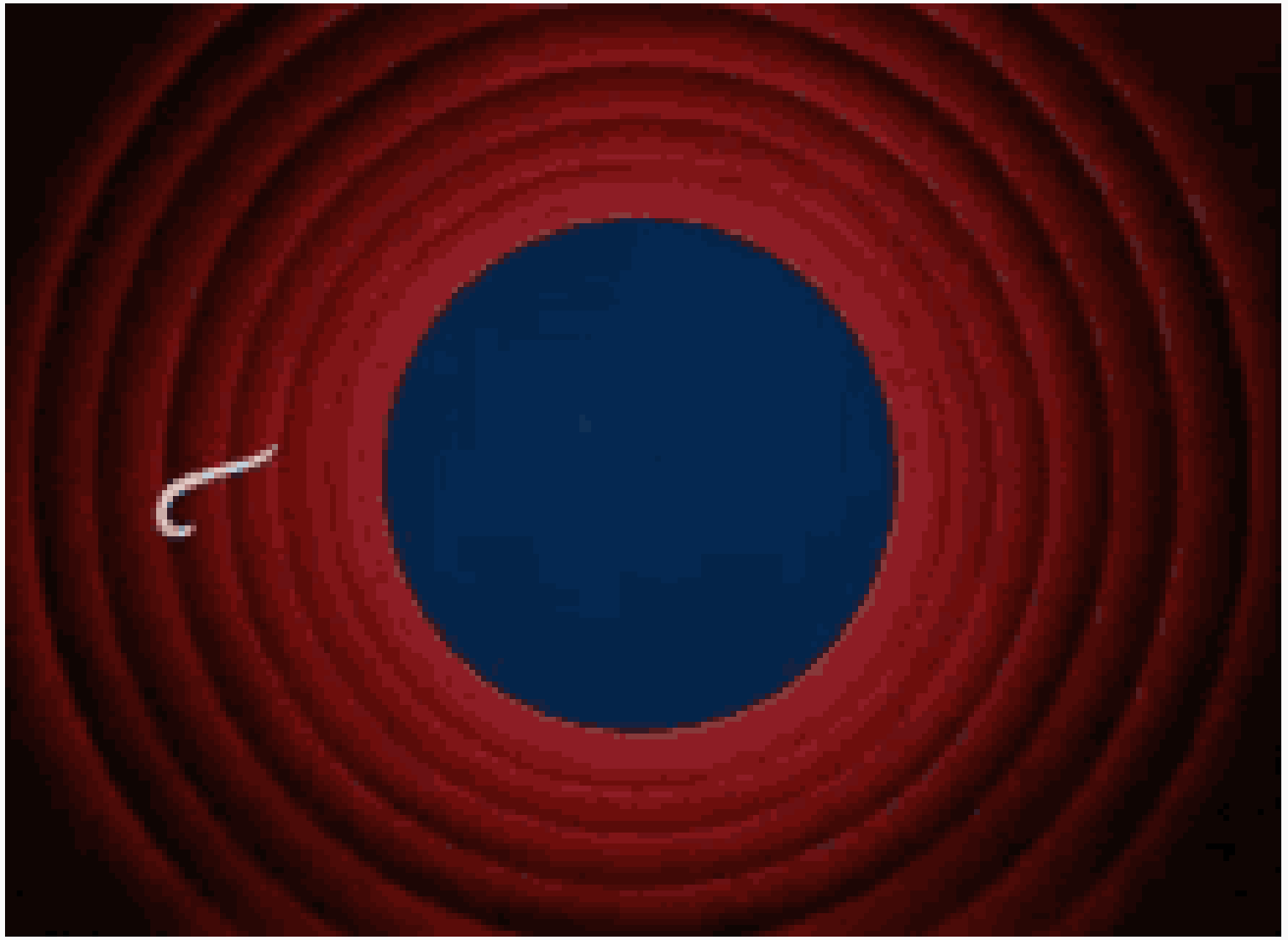




The Third Law

As a system approaches absolute zero temperature, its entropy approaches a constant minimum value

The Laws



The Laws

The First Law
Energy can neither be created nor destroyed, only converted from one form to another
The Second Law
The entropy of an isolated system always increases over time

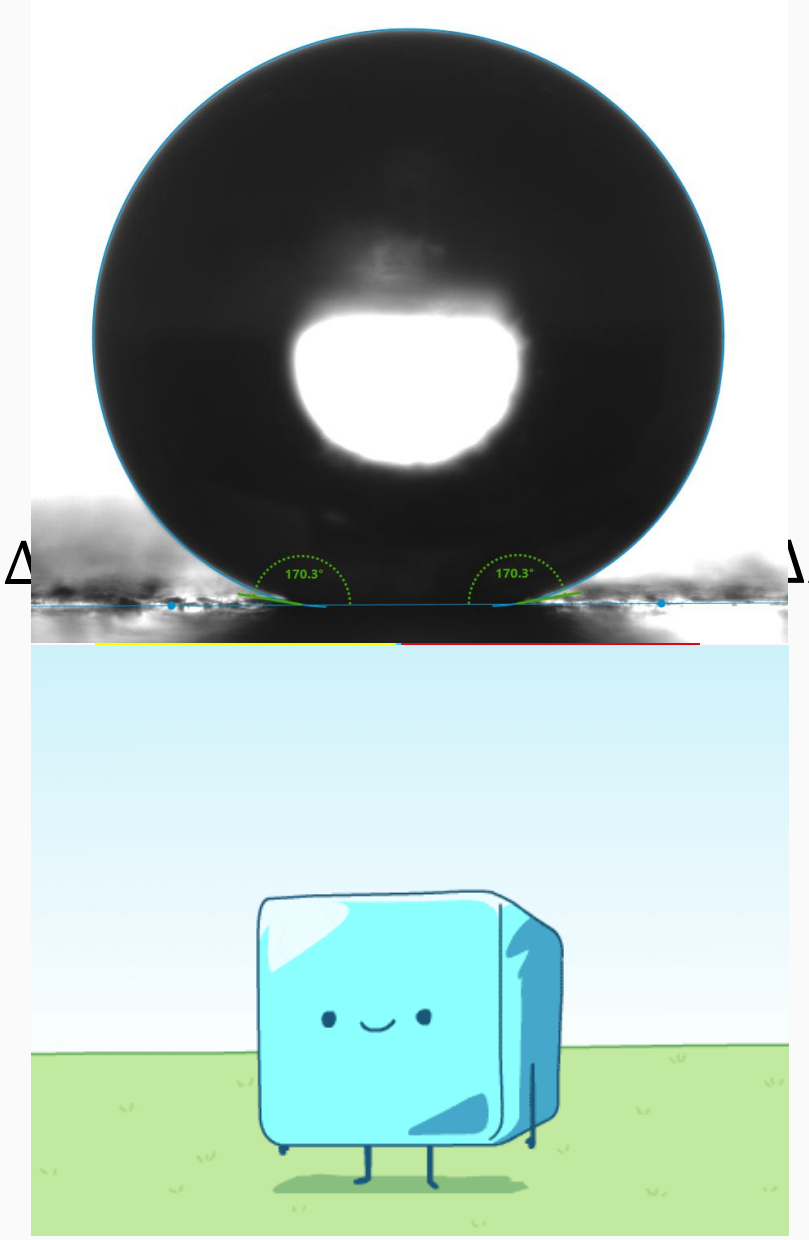
The Laws

- Entropy is a measure of disorder or randomness
- Entropy favours disorder

$$G = H - TS$$

- A reaction is spontaneous when its Gibbs free energy decreases

- Enthalpy measures the energy content of a system
- Enthalpy favours lower energy



Energy

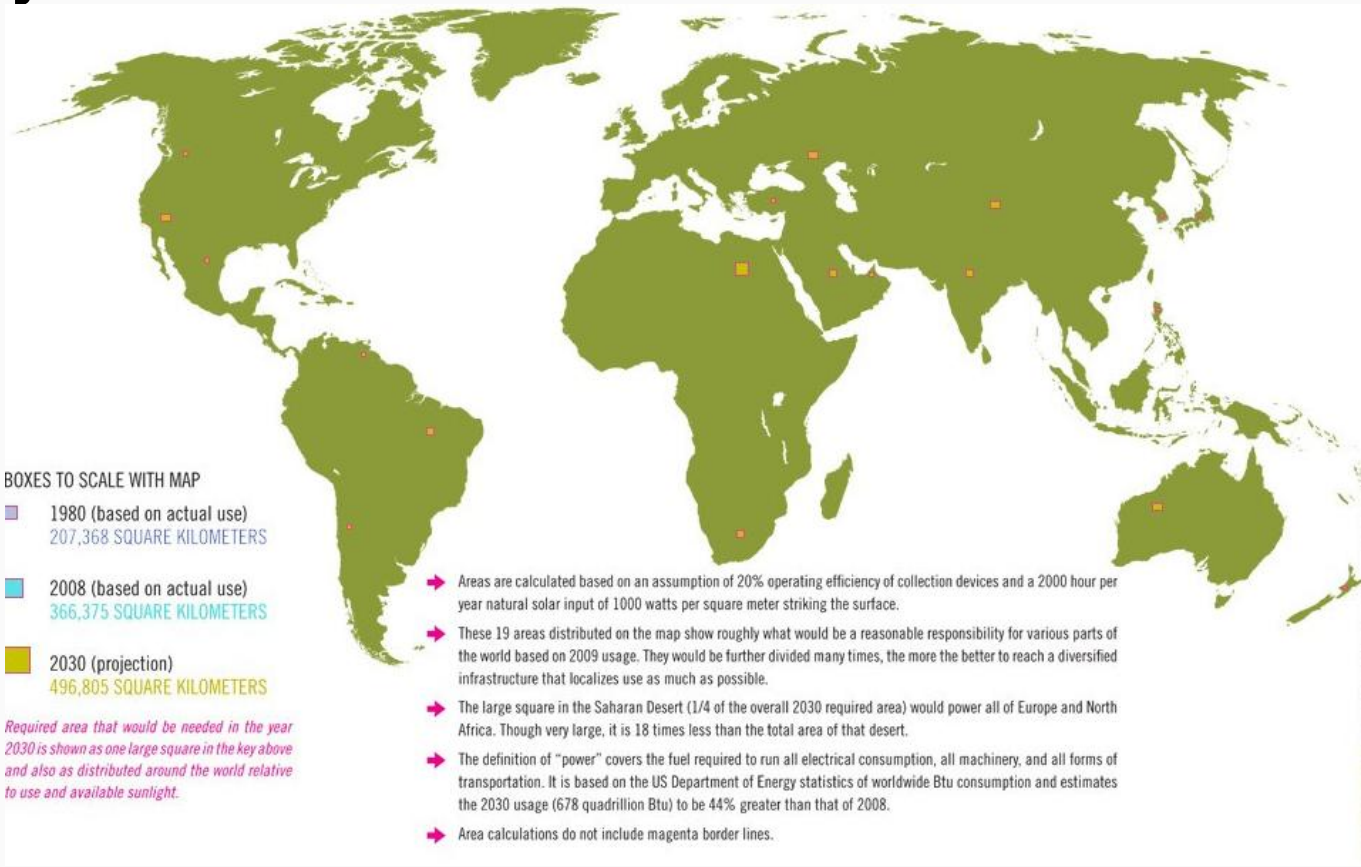
$$E = h\nu$$

The capacity to do work

Every interaction in the universe involves energy
Energy is measurable and quantifiable
Energy is governed by Thermodynamics

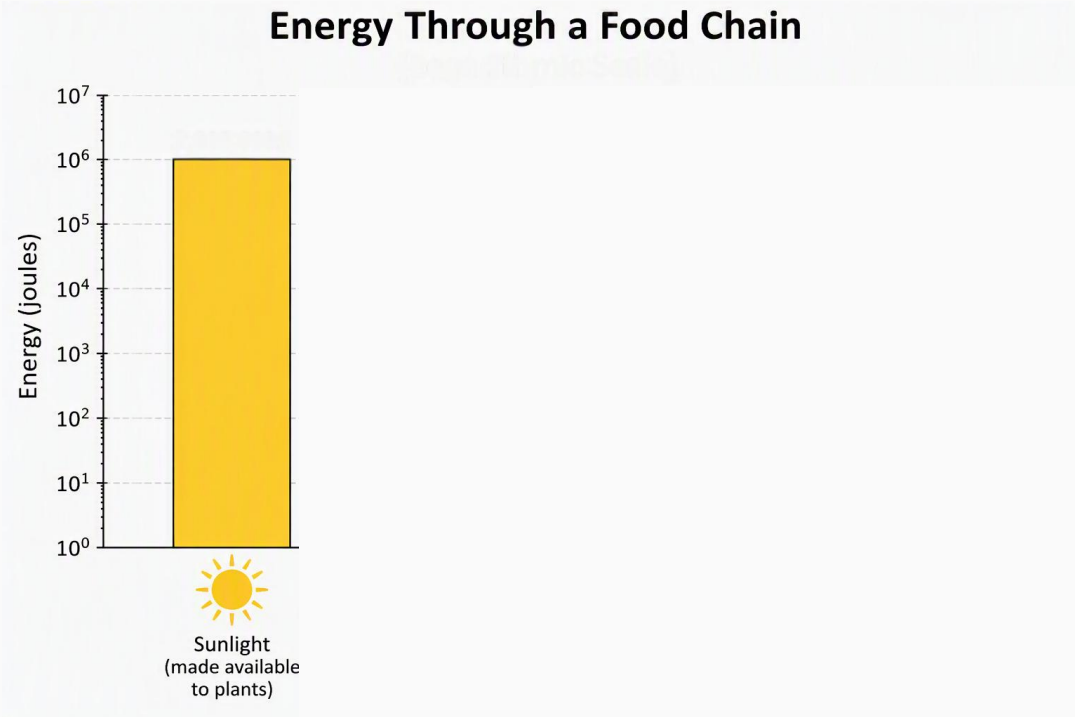
Roughly the size of Spain or France
¼ of the Saharah desert

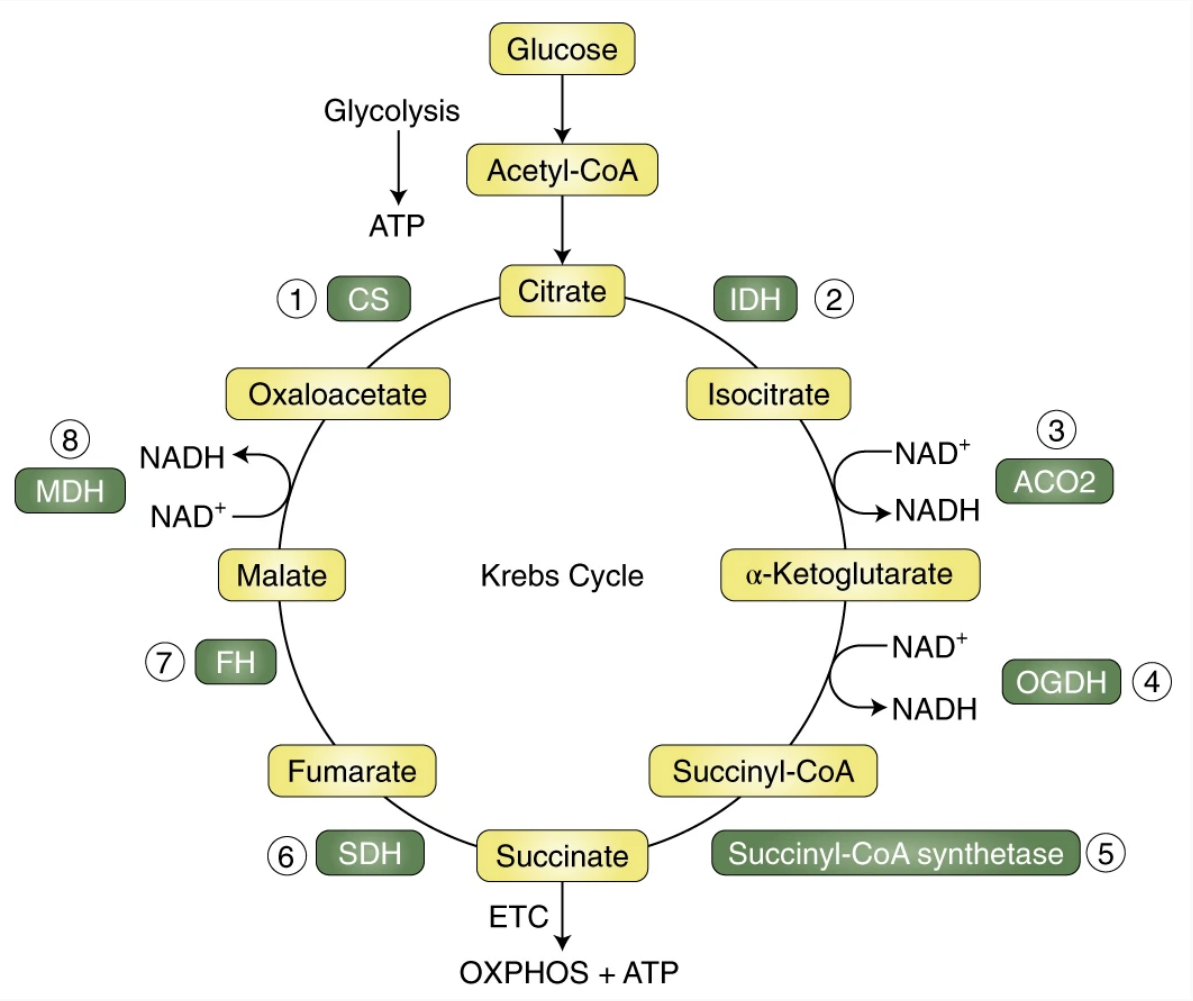
99% of energy used to sustain life comes from the Sun



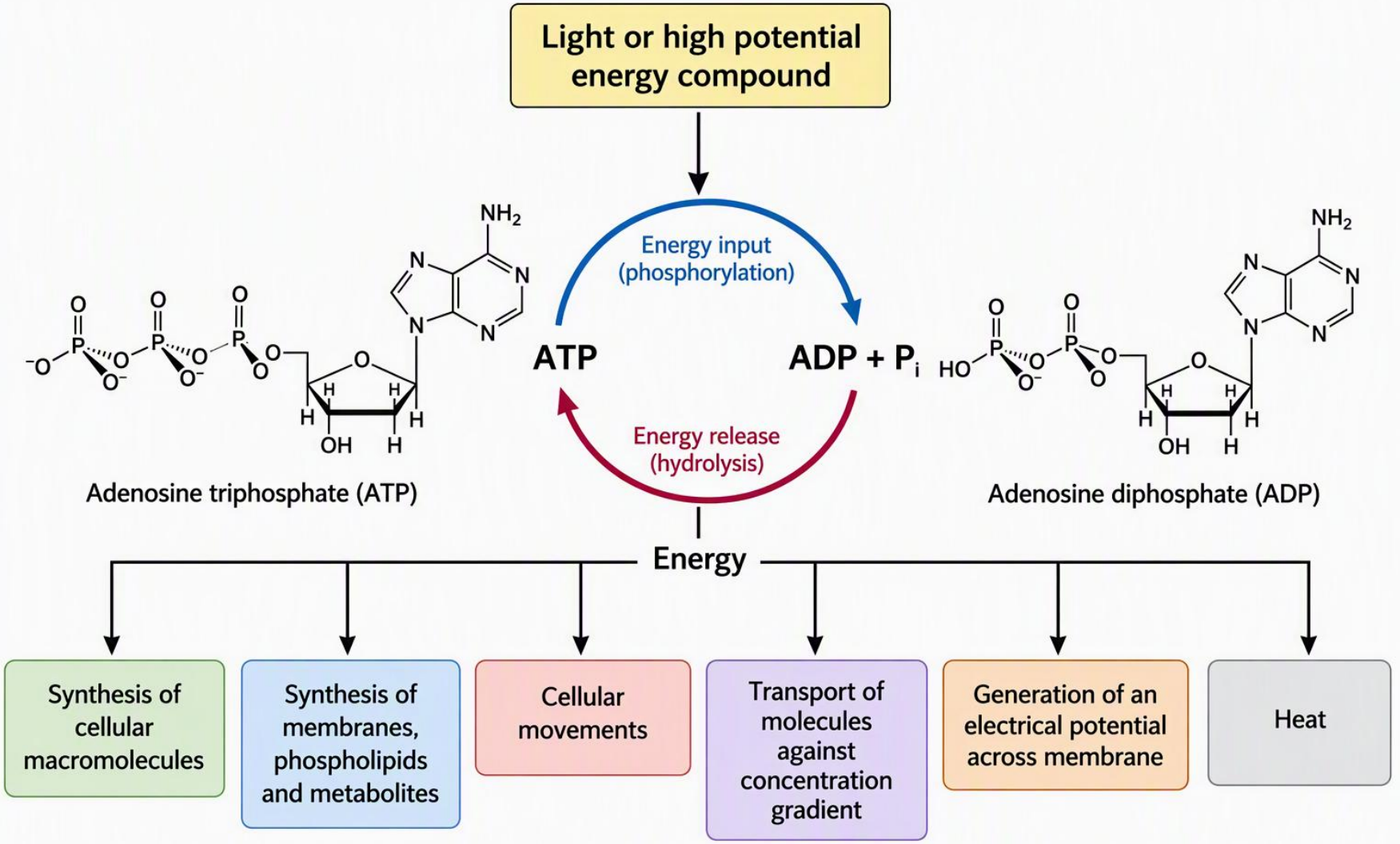
The Flow of Life

1. Energy from the Sun
2. Plants convert CO_2 and H_2O to sugars
3. Plants are eaten by animals
4. Animals are eaten by animals
5. We end up with 0.01% of the energy from the Sun





The Flow of Life



Not All Energy is Equal

- If energy can neither be created or destroyed, then why do we worry about energy?
- Different types of energy
- The greater the energy concentration the more useful the energy tends to be
- The energy gradient is important



The First Law

Science Museum London



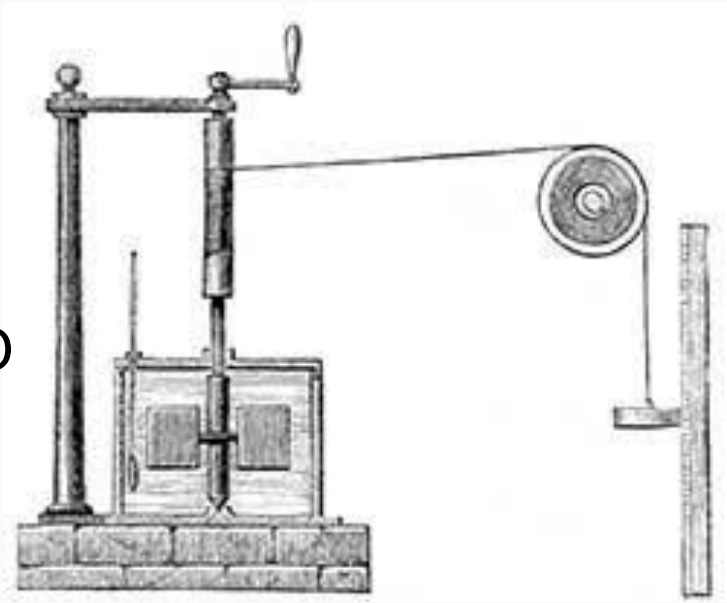
- Energy cannot be created or destroyed
- The total energy of the universe is fixed/constant
- Energy can only be transferred from one kind to another

James Prescott Joule (1818)

- Paddle wheel experiment
- Gravitational potential energy – mechanical work – heat
- Heavier weights caused greater temperature rises
- Higher drops caused greater temperature rises

$$E_p = mgh = mc\Delta T = q$$

- 1 cal is the energy required to increase the temp of 1 g H_2O by 1 K
- 1 cal is 4.2 J or 4.2 kg m^2 s^{-2}
- Heat capacity (*c*) is J kg^{-1} K^{-1}



Types of Energy

- While all energy is energy, it manifests in different ways
- Some manifestations are more useful than others

The current Super El Niño is estimated to release 10 Zeta J (10^{22}) of energy into the atmosphere, where humans use ~0.6 ZJ per annum…

Work done: $E = Fx$ (force distance)

Kinetic energy: $E = \frac{1}{2}mv^2$

Electrostatic potential: $V = \frac{q_1 q_2}{4\pi\varepsilon_0 r}$

Thermal energy: $q = mc\Delta T$

Mass energy: $e = mc^2$

Stretching a spring: $w = \frac{1}{2}kx^2$

Gravitational Energy: $E_p = mgh$

Electrical energy expended: $E = VIt$

Chemical energy: $q = n\Delta H$

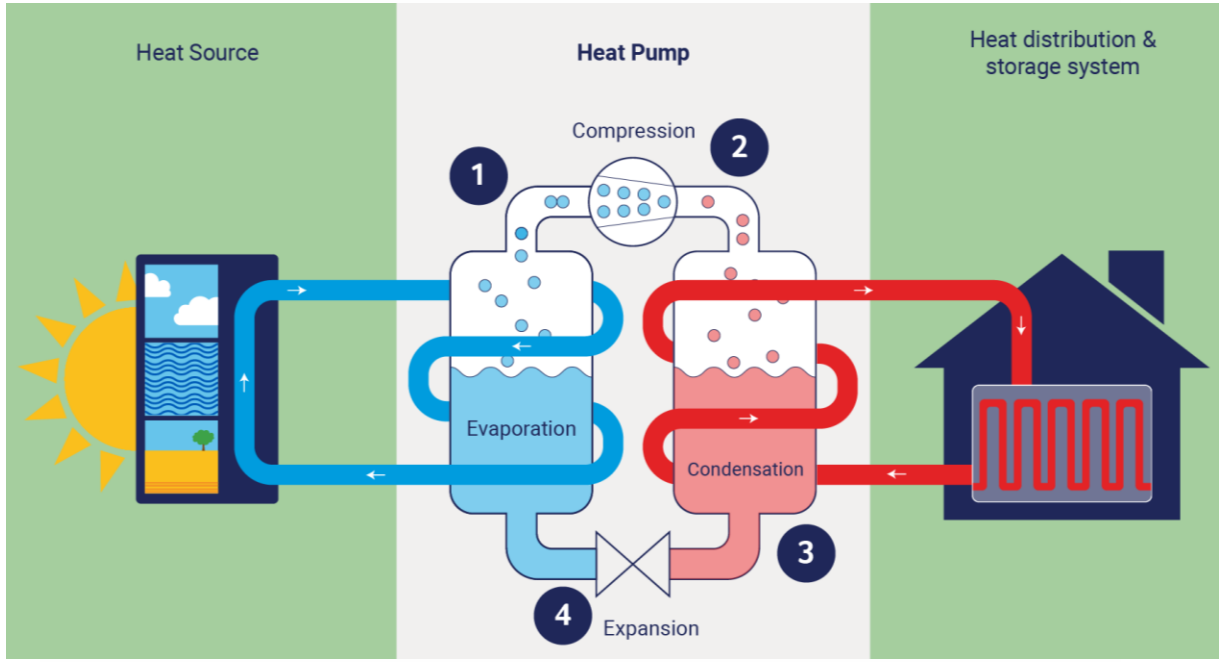
Many, many more

Measuring Energy

- How was energy first measured accurately?

Joseph Black (1728)

- 2 substances of equal mass change temperature at different rates
- All matter has a **Specific Heat Capacity**
- Heat applied to ice does not cause a rise in temperature
- All phase changes have a **Specific Latent Heat**
- Melting is endothermic
- Boiling is endothermic
- Freezing is exothermic
- Condensation is exothermic
- Heat pumps are an example of this theory applied



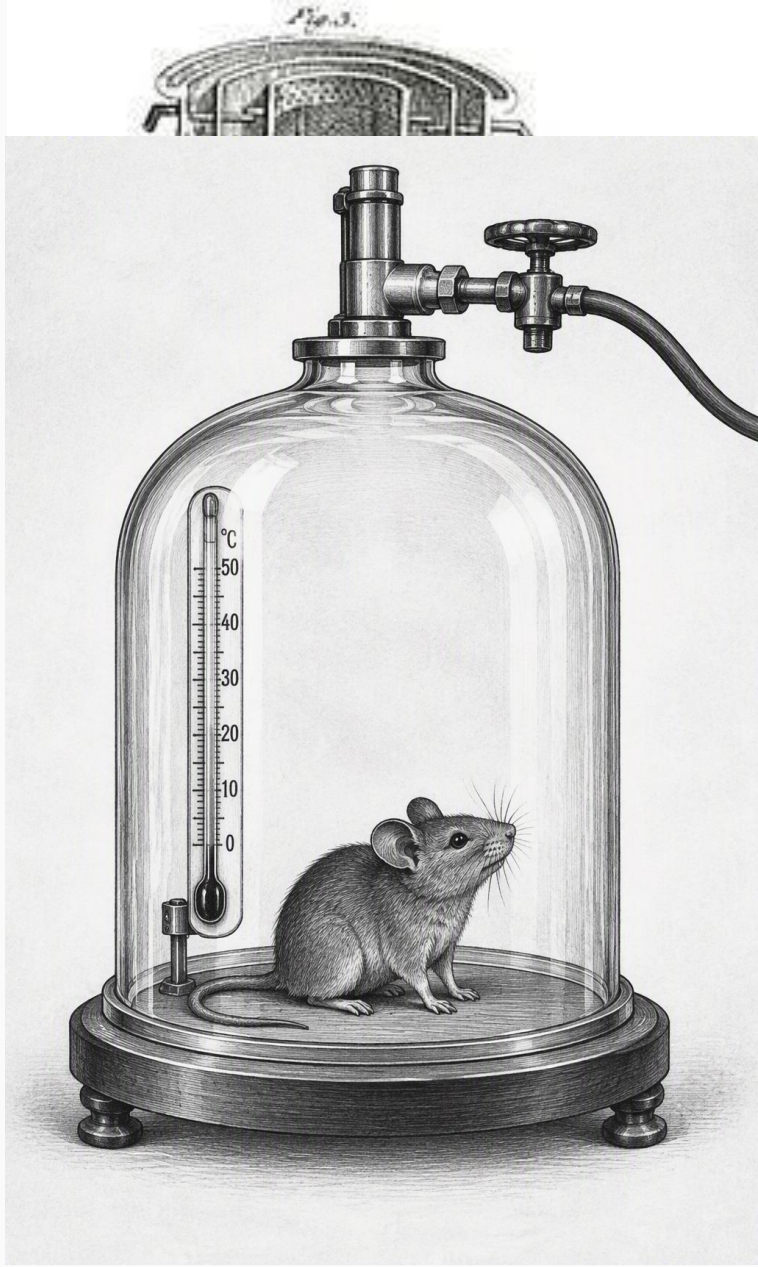
Calorimetry

- How was energy first measured accurately?

  Antoine-Laurent de Lavoisier (1743)

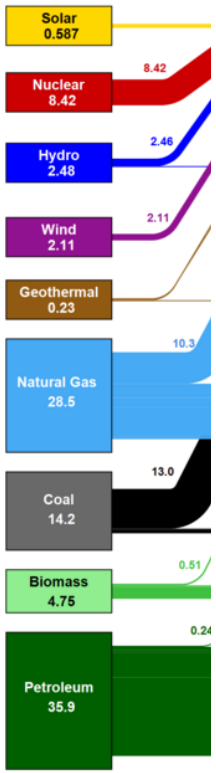
  Pierre-Simon Laplace (1749)
- Insulated device that measures energy through the melting of ice
- X mouse time = X combustions of sugar
- X mouse temp rise = X CO_2 = X sugar temp rise
- Lavoisier supported the revolution but was arrested

  Marie-Anne Pierrette Paulze Lavoisier (1758)
- Laplace helped shape France
- Lavoisier conviction deemed unjust & overturned
- How they used to determine food calories
- With food, 1 Cal = 1000 cal or 1 kcal



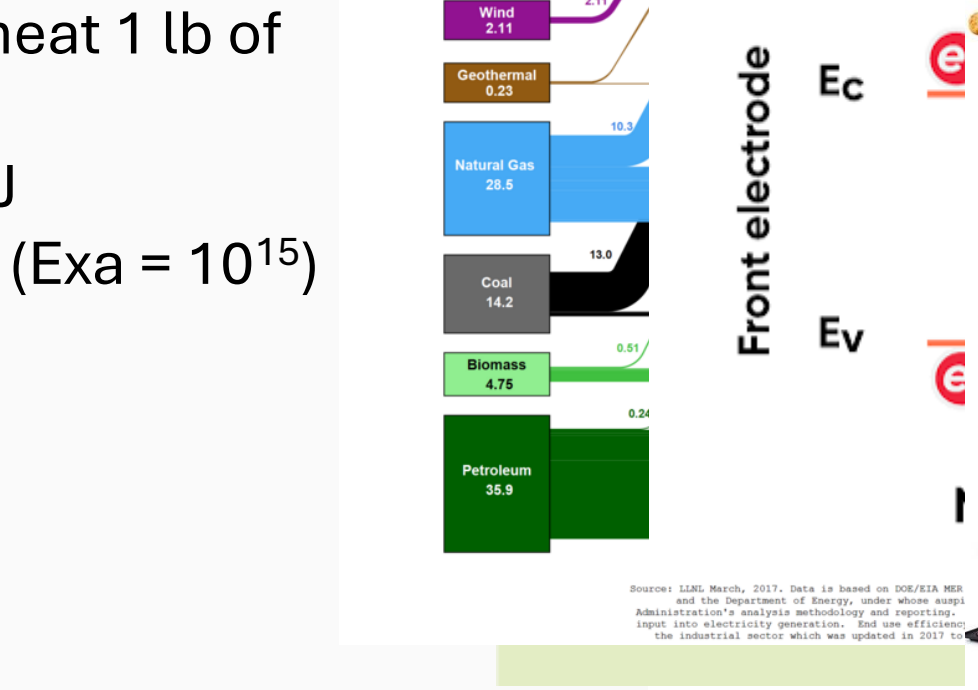
The Units of Energy

- **Joules (J) = 1 kg m^2 s^{-2}**
- cal = 4.184 J
- kWh = 3.6 x 10^6 J
- 1 eV = 1.602×10^{-19} J
- 1 BTU ≈ 1055 J (heat 1 lb of water by 1 °F)
- 1 therm ≈ 105 MJ
- 1 quad ≈ 1.05 EJ (Exa = 10^{15})



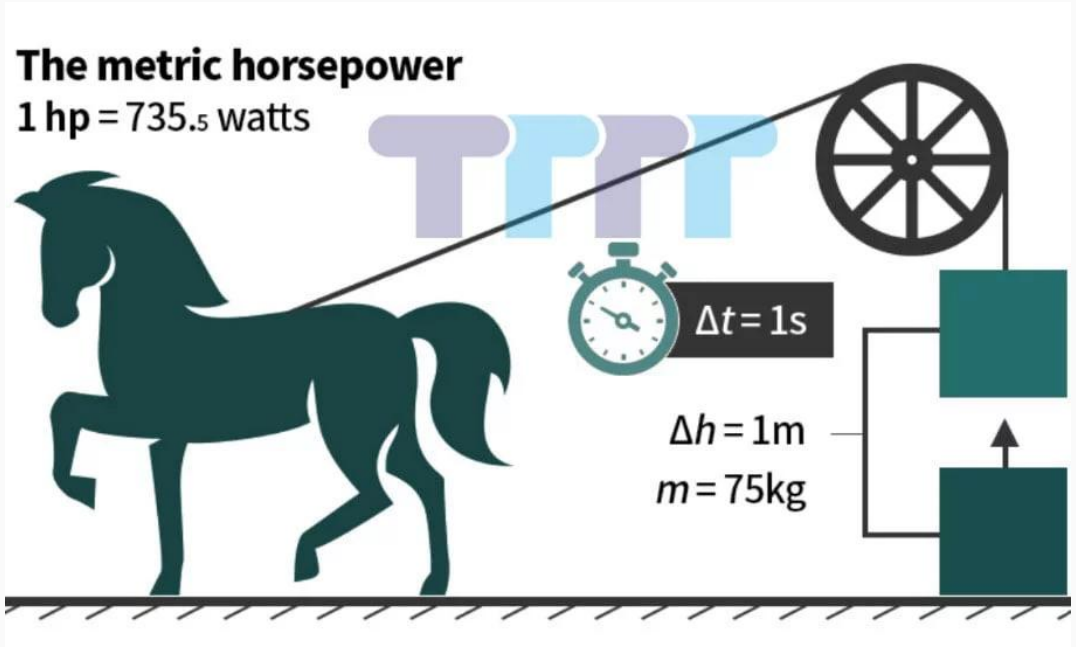


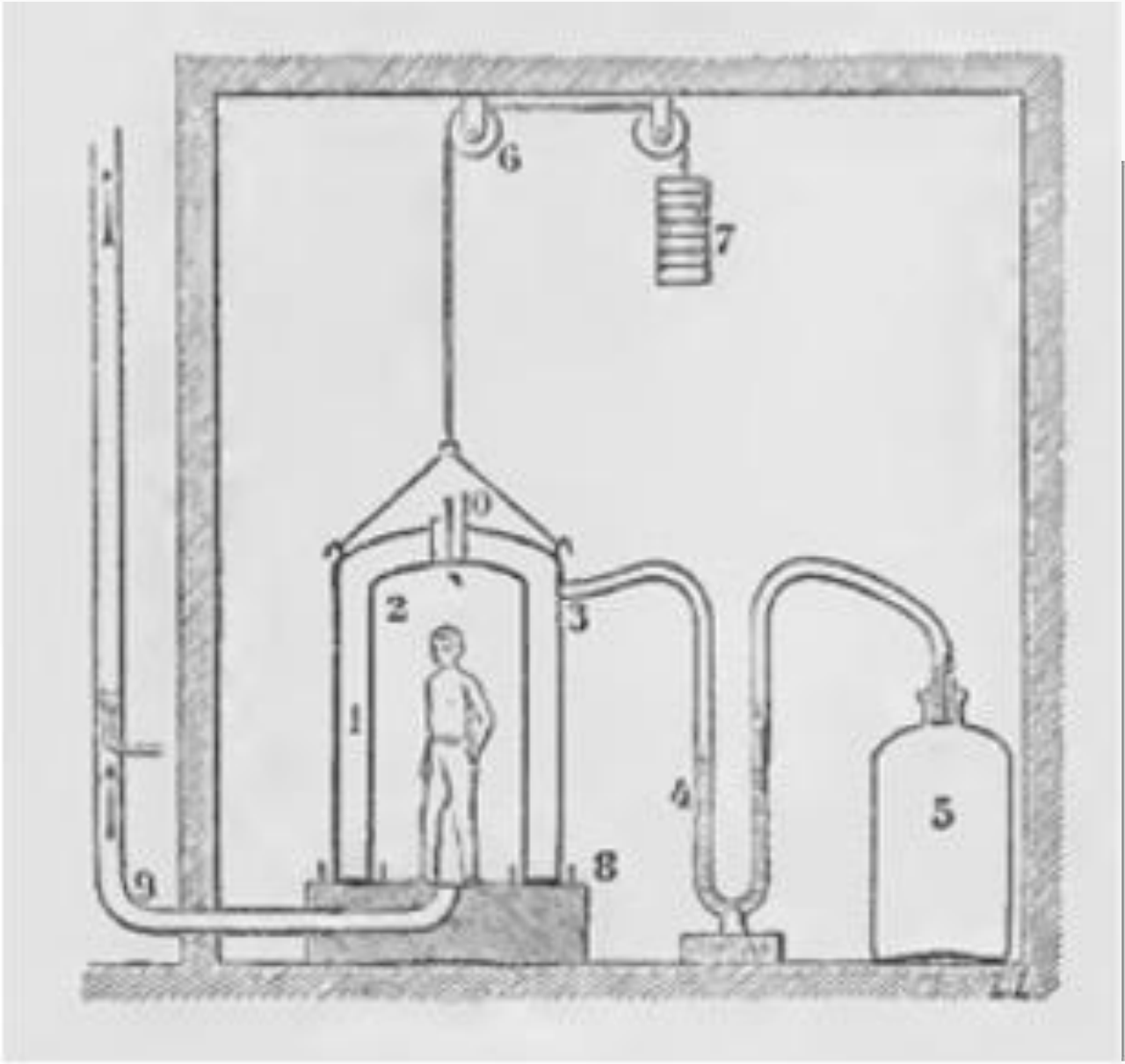




Metabolism

- Metabolic rate is power output
- $\boldsymbol{P = \frac{E}{t},\ P = \frac{J}{s},\ P = W}$
- 95% of 108 Cal = 452,000 J
- Enough to boil 1 L of water (kettle 2 kW)
- Kleiber's Law $\boldsymbol{P \approx M^{3/4}}$

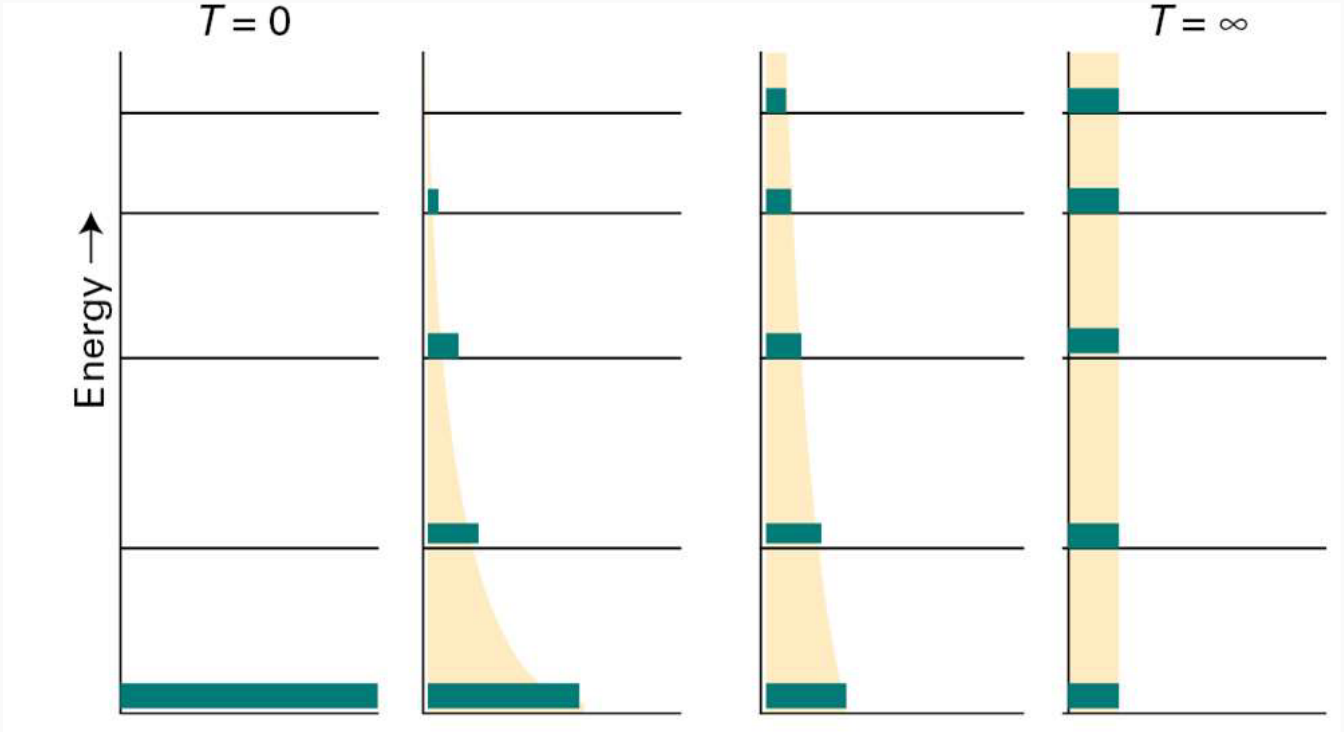




Distribution of Energy

- **Potential Energy (Stored Energy)**
  Bond Energy Chemical Energy Electrical Energy
- **Kinetic Energy (Flowing Energy)**
  Thermal Energy Radiant Energy Electrical Energy

Ludwig Boltzmann (1844)



$$\frac{N_n}{N_0} = e^{\frac{-(E_n - E_0)}{k_B T}}$$

The Boltzmann constant (k_B) is used per particle
The universal gas constant (R) is used per mole

Distribution of Energy

James Clerk Maxwell (1831)

- Maxwell derived the velocity of gases using the Boltzmann distribution
- Then integrated to give the speed distribution law for gasses

$$\vec{v} = \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} e^{\frac{-(mv^2)}{2k_B T}}$$

DO NOT NEED TO KNOW!

DO NOT NEED TO KNOW!

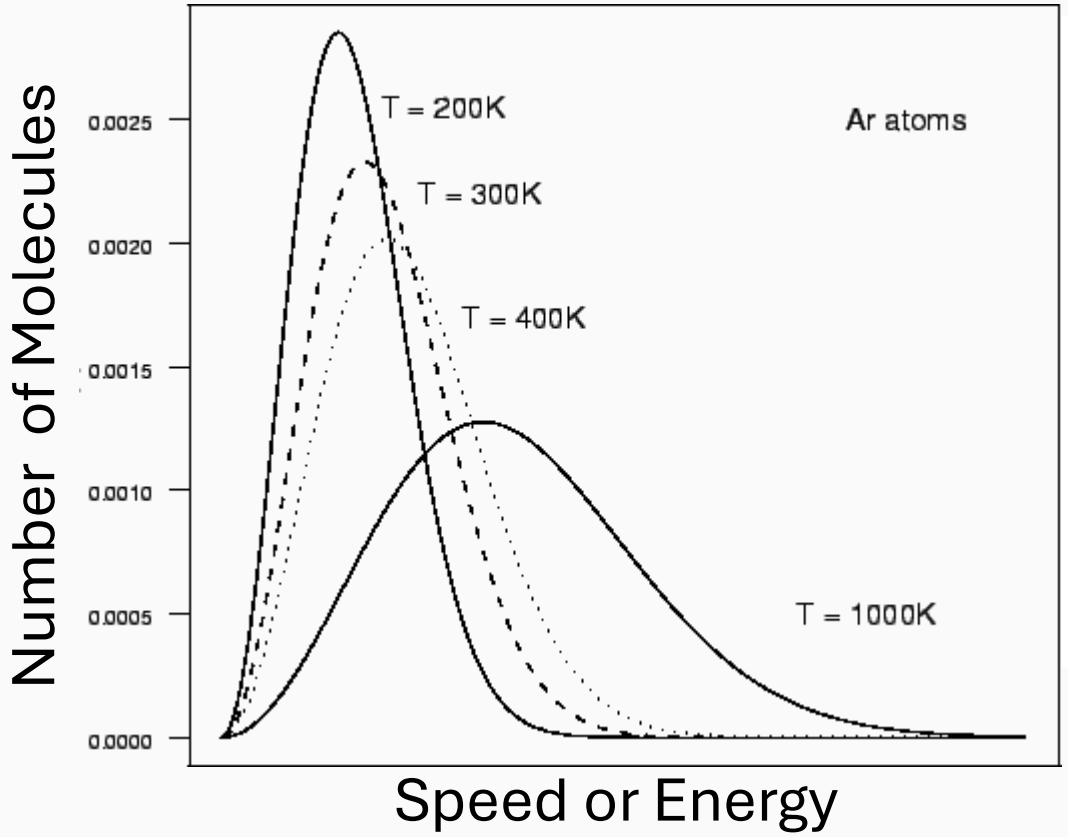
$$\int_0^{\infty} f(v) = 4\pi \left(\frac{m}{2\pi k_B T}\right)^{\frac{3}{2}} v^2 e^{\frac{-(mv^2)}{2k_B T}}$$

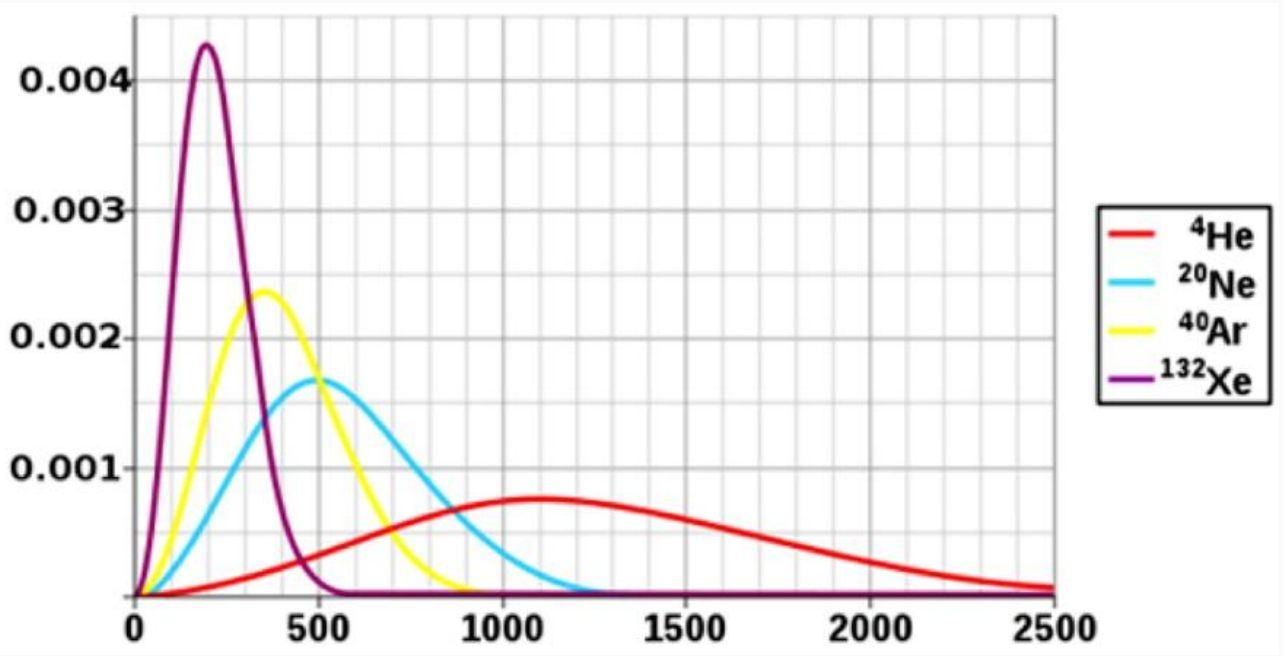
https://phet.colorado.edu/sd/simulations/gas-properties

Distribution of Energy

James Clerk Maxwell (1831)

- High speed tails explain how evaporation occurs at low temperatures
- Most particles lack the energy to escape from the surface
- Molecules in the high-speed tail do not

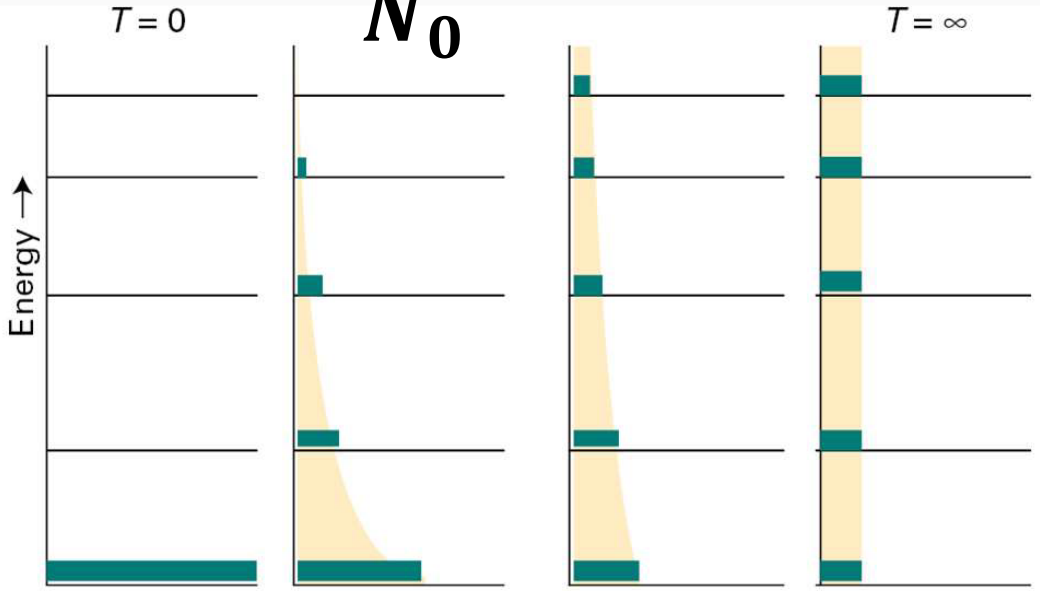




Temperature

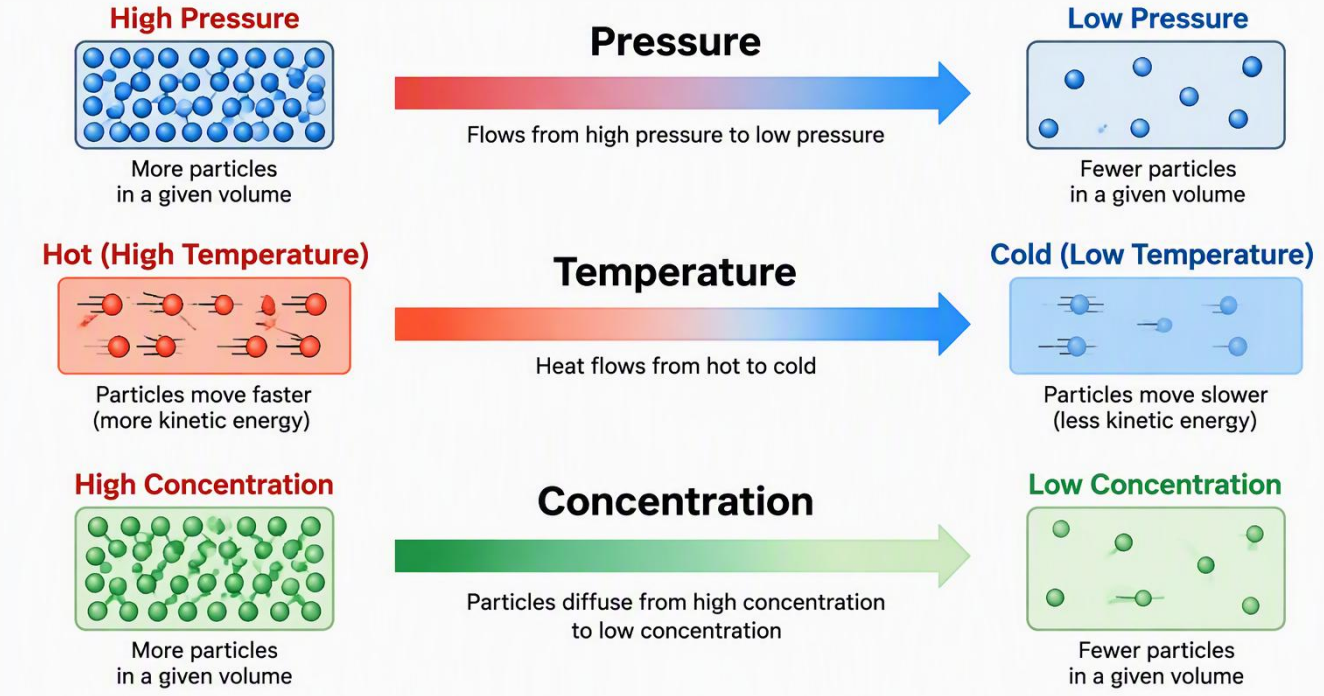
- Temperature is a directly measurable quantity that provides information about the average kinetic energy of particles in a system
- Temperature determines the distribution of energy among particles in a system, as described by the Boltzmann distribution.
- Since chemical systems possess discrete energy levels it is possible to predict the population of each
- Temperature reflects the average kinetic energy of particles and influences the distribution of energy within a system

$$\frac{N_n}{N_0} = e^{\frac{-(E_n - E_0)}{k_B T}}$$



Temperature

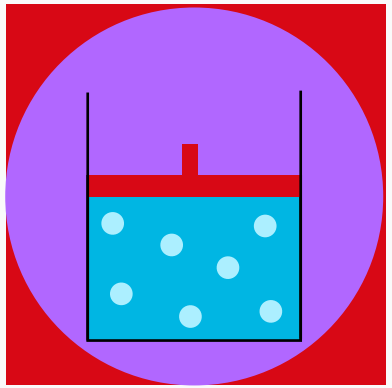
- What does temperature tell us?
- Temperature tells us the direction of energy flow
- Systems move toward lower-energy states

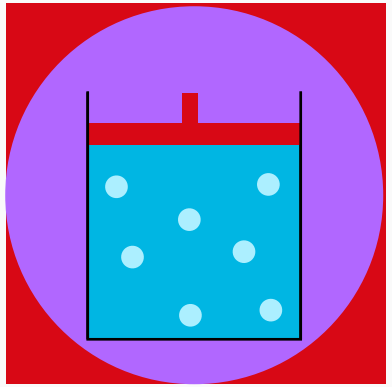


- The total energy doesn't change
- **Intensive property**, not additive, not dependant on the amount
- Splitting does not change it

The System, Surroundings, Universe

- **The Universe** surrounds both the system and the surrounding
- **The Surroundings** is everything around the system that has physical contact with it and may or may not exchange energy & matter with it
- **The System** the source under consideration
- Open system can exchange energy and matter with surroundings
- Closed can exchange energy with surroundings
- Isolated does not exchange anything with the surroundings
- Biological systems are open systems
- Plants take light, minerals, water, CO_2 from their surroundings
- Animals exchange matter by eating, breathing, defecating, etc…











Quantifying the First Law

- The internal energy **U** is the sum total of all possible energies in the system
- Electronic, vibrational, kinetic, bond energy, etc…
- **The absolute value of U <u>cannot</u> be determined**
- U is an **extensive property** it is dependant on how much of it you have
- Internal energy can be changed by transferring energy in or out of the system

$$\Delta U = q + w$$

- w & q are boundary phenomena, moving energy in and out of a system
- U is a **State Function**
- It doesn't matter how you got there





Energy Transfer

- **Heat** is the transfer of energy **due to a difference in temperature** between the system and surroundings, resulting in a change in the **random** motion of particles
- **Work** is the transfer of energy to or from the system that results in a **coherent/coordinated** motion of the molecules of a system
- Neither heat nor work are a form of energy, they are energy transfer process
- It is not always obvious what process is occurring



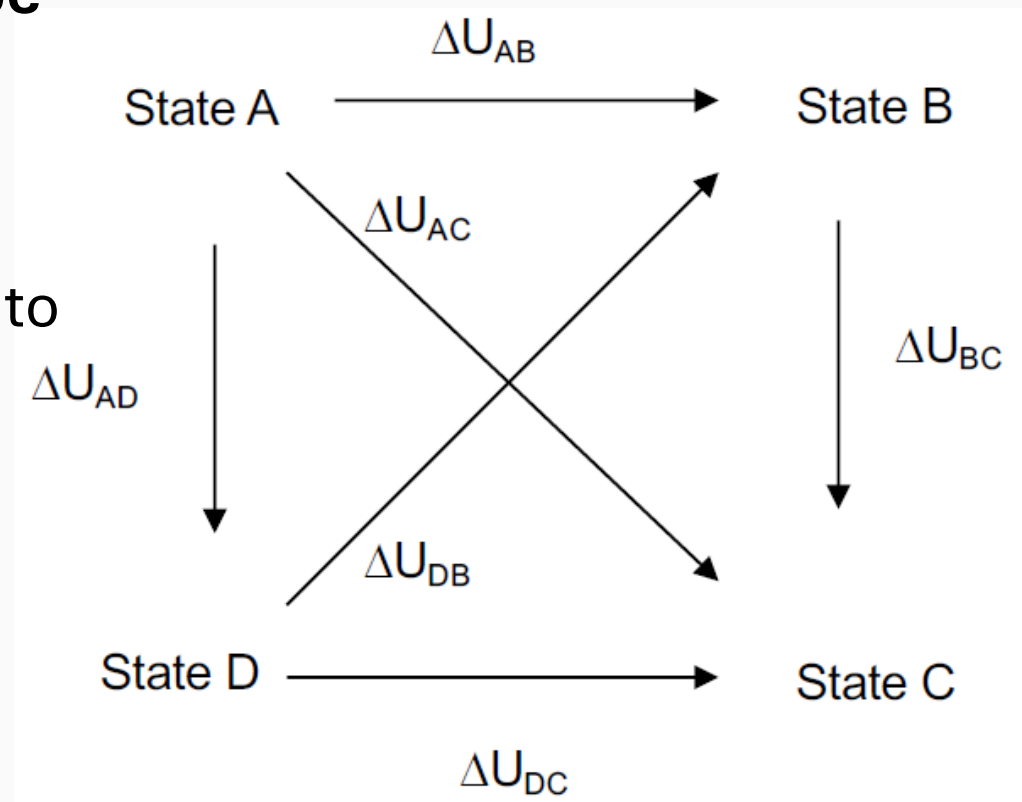


Endothermic vs Exothermic

- **The change of any property is always taken to be with respect to the system**

$$\Delta U = q + w$$

- **Endothermic**
  - ***q*>0** energy is moving from the surroundings to the system
  - ***w*>0** surroundings do work on the system
- **Exothermic**
  - ***q*<0** energy is moving from the system to the surroundings
  - ***w*<0** system does work on the surroundings



$$\Delta U_{AD} + \Delta U_{DC} = \textbf{State C}$$

$$(-30 + (-20)) + (-30 + (-20)) = -20 kJ\ mol^{-1}$$

$$\Delta U_{AC} = \textbf{State C}, -50 + (-30) = -20 kJ\ mol^{-1}$$

Path Functions

- ***w* & *q* are path-dependent because the amount of heat and work transferred depends on how the process occurs, not just on the initial and final states**
- Calculate the work done by a gas

$$w = -FD, \qquad w = -p_{ext}AD, \qquad w = -p_{ext}\Delta V$$

- **Enthalpy**

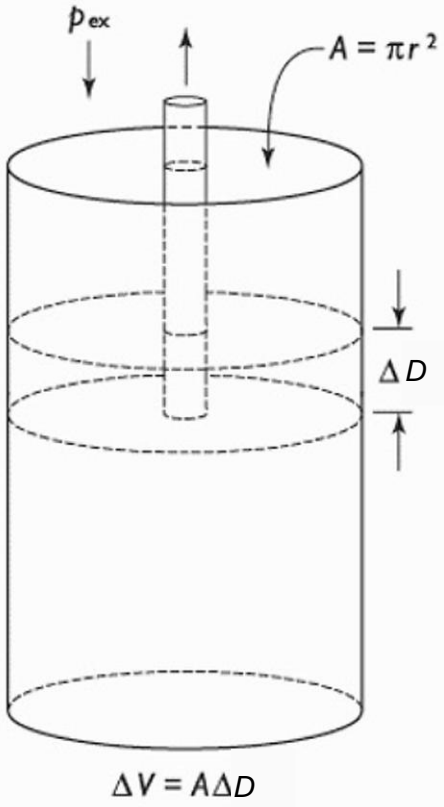
$$\Delta H = \Delta U + p\Delta V, \qquad \Delta H = \Delta U - w$$

- At constant pressure

$$\Delta U = q_p + w, \qquad q_p = \Delta U - w, \qquad q_p = \Delta H$$

- At constant volume

$$\Delta H = \Delta U + p\Delta V, \qquad \Delta H = \Delta U$$



Enthalpy

$$\Delta H = \Delta U + p\Delta V, \qquad \Delta H = \Delta U, \qquad q_p = \Delta H$$

- **Enthalpy is a state function**
- It is the internal energy of a system plus the energy needed to make room for that system by displacing its surroundings
- Enthalpy changes can be approximated as the energy changes arising from differences in bond strengths of starting (reactants) and end points (products)
- $\Delta H > 0$ energy is moving from the surroundings into the system (**Endothermic**)
- $\Delta H < 0$ energy is moving from the system into the surroundings (**Exothermic**)
- Measuring enthalpy can give us the internal energy at constant volume

Enthalpy of Formation & Hess's Law

- Enthalpy of formation is the enthalpy change when one mole of a compound is formed from its constituent elements in their standard states under standard conditions
- Hess's Law states the enthalpy change for a reaction is independent of the route taken between the initial and final states

$$C_{(s)} + O_{2(g)} = CO_{2(g)}, \qquad C_{(s)} + \tfrac{1}{2}O_{2(g)} = CO_{(g)}, \qquad CO_{(g)} + \tfrac{1}{2}O_{2(g)} = CO_{2(g)}$$

$$\Delta H_{overall} = \Delta H_1 + \Delta H_2$$

O=O 498 kJ mol^{-1} C≡O 1072 kJ mol^{-1} C=O 805 kJ mol^{-1} C_s 717 kJ mol^{-1}

$$\Delta H = 717 + 498 - 805 \text{ x } 2 = -395 \text{ kJ mol}^{-1}$$

or

$$717 + 249 - 1072 = -106 \text{ kJ mol}^{-1}$$

+

$$1072 + 249 - 805 \text{ x } 2 = -289 \text{ kJ mol}^{-1}$$

Hess's Law Calculation

- Lactic acid fermentation using glucose can be summarized as follows:

$$\mathbf{C_6H_{12}O_6 \rightarrow 2CH_3COHCOOH}$$

- Calculate the enthalpy change for lactic acid fermentation

$$\boldsymbol{\Delta H_c\ Glucose = -2805\ kJ\ mol^{-1}}, \qquad \boldsymbol{\Delta H_c\ Lactic\ Acid = -1364\ kJ\ mol^{-1}}$$

$$\mathbf{C_6H_{12}O_6 + 6O_2 \rightarrow = 6CO_2 + 6H_2O = -2805\ kJ\ mol^{-1}}$$

$$\mathbf{2CH_3COHCOOH + 3O_2 \rightarrow 6CO_2 + 6H_2O = -2728\ kJ\ mol^{-1}}$$

$$\boldsymbol{\Delta H = -2805 - (-2728) = -77\ kJ\ mol^{-1}}$$

Bond Enthalpies

- Mean bond enthalpies (refenced for the gas phase) allow you to predict formation enthalpies for molecules, or even enthalpies of reaction.

Common Average Bond Enthalpies

(Average values, in kJ mol^{-1})

Hydrogen and Halogens

Bond	ΔH_B / kJ mol^{-1}
H–H	436
H–F	565
H–Cl	432
H–Br	366
H–I	299
F–F	158
Cl–Cl	243
Br–Br	193
I–I	151

Carbon Bonds

Bond	ΔH_B / kJ mol^{-1}
C–H	413
C–C	347
C=C	612
C≡C	837
C–O	358
C=O (aldehyde/ketone)	743
C=O (CO_2)	805
C–N	305
C≡N	891
C–S	259

Other Important Bonds

Bond	ΔH_B / kJ mol^{-1}
O–H	463
N–H	391
O=O	498
N≡N	945
S–H	347
S=O	523
P–H	320
P=O	544
C–F	485
C–Cl	339
C–Br	285
C–I	213

$$\Delta H_{\text{reaction}} = \sum \Delta H_B \text{ (bonds broken)} - \sum \Delta H_B \text{ (bonds formed)}$$

Values are average bond enthalpies and can vary slightly depending on the source.